

Solutions to Problem Set 5

1. (a)

$$F(x) = \int_0^1 \sqrt{x^2 + t^4} dt$$

$$F'(x) = \int_0^1 \frac{\partial}{\partial x} \sqrt{x^2 + t^4} dt = \int_0^1 \frac{x}{\sqrt{x^2 + t^4}} dt$$

(b)

$$F(x) = \int_{x^{\frac{1}{3}}}^1 e^{-xt^3} dt$$

$$\begin{aligned} F'(x) &= -e^{-x^2} \frac{d}{dx} x^{\frac{1}{3}} + \int_{x^{\frac{1}{3}}}^1 \frac{\partial}{\partial x} e^{-xt^3} dt \\ &= -\frac{1}{3} e^{-x^2} x^{-\frac{2}{3}} - \int_{x^{\frac{1}{3}}}^1 t^3 e^{-xt^3} dt \end{aligned}$$

2. (a) The equation is separable. We have

$$\int \frac{dy}{y^2} = \int dt + C,$$

so that

$$\frac{-1}{y} = t + C.$$

Using the given initial condition we find that

$$-2 + 0 + C,$$

so that $C = -2$. We conclude that

$$y = \frac{1}{2-t}.$$

(b) The equation is linear. An integrating factor is given by $\mu(t) = e^{-2t}$. Multiplying the differential equation by μ we obtain

$$e^{2t}y'(t) - 2e^{-2t}y(t) = te^{-2t},$$

which is equivalent to

$$\frac{d}{dt}(e^{-2t}y(t)) = te^{-2t}.$$

Using integration by parts on the right-hand side, we find that

$$e^{-2t}y(t) = -\frac{1}{2}te^{-2t} - \frac{1}{4}e^{-2t} + C.$$

Putting $t = 0$ and using the initial condition, we find that $C = \frac{13}{4}$. Multiplying through by e^{2t} , we obtain

$$y(t) = -\frac{1}{2}t - \frac{1}{4} + \frac{13}{4}e^{2t}.$$

- (c) The equation is linear. An integrating factor is given by $\mu(t) = e^{t^2}$. Multiplying the differential equation by μ we obtain

$$e^{t^2}y'(t) + 2te^{t^2}y(t) = 1,$$

which is equivalent to

$$\frac{d}{dt}(e^{t^2}y(t)) = 1.$$

Integrating this equation, we find that

$$e^{t^2}y(t) = t + C.$$

Putting $t = 0$ and using the initial condition, we find that $C = -1$. Multiplying through by e^{-t^2} we obtain

$$y(t) = (t - 1)e^{-t^2}.$$

- (d) The equation is linear. An integrating factor is given by

$$\mu(t) = \exp[-5 \ln t] = t^{-5}.$$

Multiplying the differential equation by μ we obtain

$$t^{-5}y'(t) - 5t^{-6}y(t) = t^5,$$

which is equivalent to

$$\frac{d}{dt}(t^{-5}y(t)) = t^5.$$

Integrating this equation, we find that

$$t^{-5}y(t) = \frac{1}{6}t^6 + C.$$

Putting $t = 1$ and using the initial condition we find that $C = \frac{41}{6}$. Multiplying through by t^5 we obtain

$$y(t) = \frac{1}{6}t^{11} + \frac{41}{6}t^5.$$

(e) The equation is separable. We have

$$\int \frac{dy}{1+y} = \int t^2 dt + C,$$

so that

$$\ln(1+y) = \frac{1}{3}t^3 + C.$$

Using the given initial condition, we find that

$$\ln 2 = C,$$

so that

$$\ln(1+y) = \frac{1}{3}t^3 + \ln 2.$$

We conclude that

$$y(t) = 2e^{\frac{1}{3}t^3} - 1.$$

3. Differentiating the integral equation, we obtain the differential equation

$$y'(t) + 4y(t) = 1.$$

The integral equation also implies the initial condition $y(0) = 3$ (by setting $t = 0$). An integrating factor is given by $\mu(t) = e^{4t}$. Multiplying the differential equation by μ we obtain

$$e^{4t}y'(t) + 4e^{4t}y(t) = e^{4t},$$

which is equivalent to

$$\frac{d}{dt}(e^{4t}y(t)) = e^{4t}.$$

Integrating this equation we obtain

$$e^{4t}y(t) = \frac{1}{4}e^{4t} + C.$$

Putting $t = 0$ and using the initial condition, we find that $C = \frac{11}{4}$. Multiplying through by e^{-4t} we obtain

$$y(t) = \frac{1}{4} + \frac{11}{4}e^{-4t}.$$

4. Dividing the differential equation by $y(t)^3$ we obtain

$$y(t)^{-3}y'(t) + y(t)^{-2} = 1.$$

With the substitution $u(t) = y(t)^{-2}$, we have

$$u'(t) = -2y(t)^{-3}y'(t),$$

so that

$$y(t)^{-3}y'(t) = -\frac{1}{2}u'(t).$$

The differential equation becomes

$$-\frac{1}{2}u'(t) + u(t) = 1,$$

which is equivalent to

$$u'(t) - 2u(t) = -2.$$

This differential equation is linear with integrating factor $\mu(t) = e^{-2t}$. The initial condition for u is $u(0) = \frac{1}{4}$. Solving for u , we obtain

$$e^{-2t}u(t) = e^{-2t} - \frac{3}{4},$$

which gives $u(t) = 1 - \frac{3}{4}e^{2t}$ and consequently

$$y(t) = \frac{1}{\sqrt{1 - \frac{3}{4}e^{2t}}}.$$

We take the positive root because $y(0) > 0$.

5. (a) The characteristic equation is

$$\lambda^2 - \lambda + \frac{1}{4} = 0,$$

and this factors to

$$\left(\lambda - \frac{1}{2}\right)^2 = 0,$$

so that $\lambda = \frac{1}{2}$ is a double root. Therefore the general solution is

$$y(t) = C_1e^{\frac{1}{2}t} + C_2te^{\frac{1}{2}t}.$$

(b) The characteristic equation is

$$\lambda^2 + 2\lambda - 15 = 0,$$

and this factors to

$$(\lambda - 3)(\lambda + 5) = 0,$$

so that the roots are $\lambda = 3, -5$. Therefore the general solution is

$$y(t) = C_1e^{3t} + C_2e^{-5t}.$$

(c) The characteristic equation is

$$\lambda(\lambda - 1) + 3\lambda + 1 = 0,$$

and this is the same as

$$\lambda^2 + 2\lambda + 1 = 0,$$

and this factors to

$$(\lambda + 1)^2 = 0,$$

so that $\lambda = -1$ is a double root. Therefore the general solution is

$$y(t) = C_1 t^{-1} + C_2 t^{-1} \ln t.$$

6. Putting in the numbers for r and σ , the differential equation becomes

$$.02x^2v''(x) + .02xv'(x) - .02v(x) = 0.$$

This is the same as

$$x^2v''(x) + xv'(x) - v(x) = 0.$$

We can solve this using the procedure in the notes or we can simply plug in $v(x) = x^\lambda$ and try to find 2 value of λ that work. Substituting this form of x into the differential equation gives

$$(\lambda(\lambda - 1) + \lambda - 1)x^2 = 0, \quad x > 0.$$

This is equivalent to $\lambda^2 - 1 = 0$, so we find

$$\lambda_1 = 1, \quad \lambda_2 = -1.$$

The general solution of the differential equation is therefore given by

$$v(x) = C_1x + C_2x^{-1}.$$

To find C_1 and C_2 , we use the given boundary conditions:

$$2,800C_1 + (2,800)^{-1}C_2 = 0, \quad 3,500C_1 + (3,500)^{-1}C_2 = 100.$$

Solving for C_1 and C_2 we find that

$$C_1 = \frac{5}{63}, \quad C_2 = -(2,800)^2 \frac{5}{63}.$$

We find that

$$v(3,100) = 45.3149.$$